

10000

Total Employees

2247

Total Attrition

7753

Active Employees

22.47%

Attrition_Rate

40

Average_Age

Job Level

1 2 3 4 5 6 7 8

286 255 263 272 306 293 276 296

Job Involvement

1 2 3 4 5

654 632 330 309 322

Work life Balance

1 2 3 4 5

604 628 375 320 320

Job Satisfaction

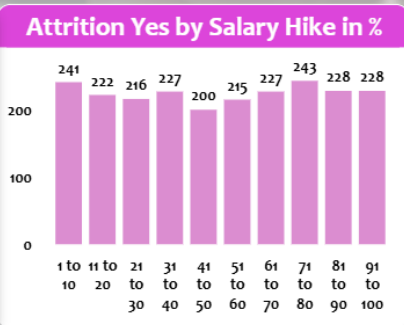
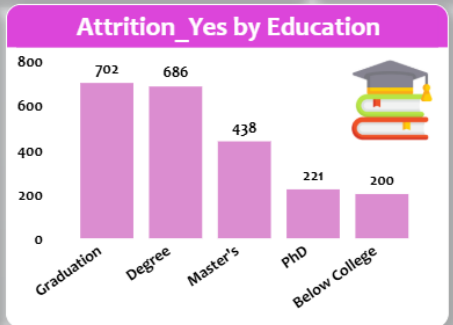
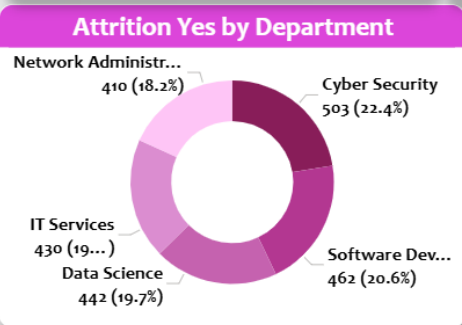
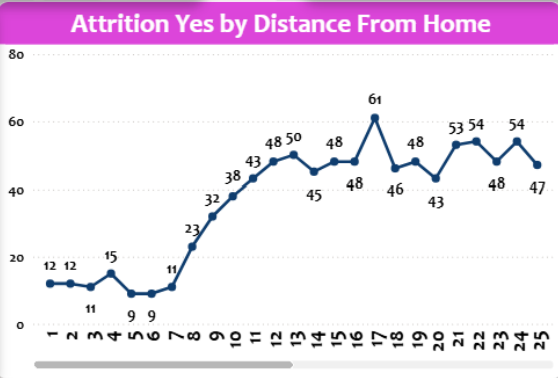
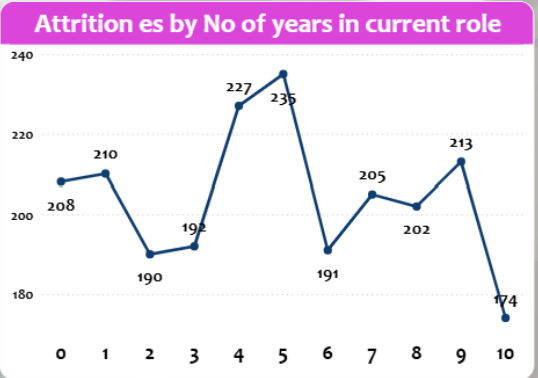
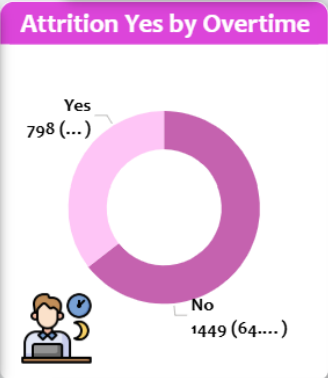
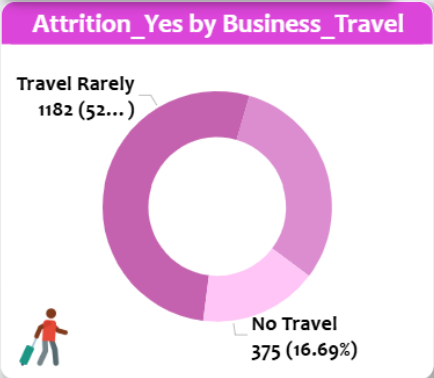
1 2 3 4 5

545 541 402 397 362

Job Environment

1 2 3 4 5

604 652 535 234 222



Attrition Rate by Job Role

Consultant	IT	Deve...	Help ...	HR
Director	Software ...			
QA Analyst	Business A...	Technician		Supp...
		Manager		



Employee Attrition Analysis

HR Analytics Presentation — **Tech IT Solutions**

SQL · POWER BI

Noor Mohammad · 15th Jun 2026

Overview



Tech IT Solutions is a service-based company providing IT and business support across finance, healthcare, and retail. With ~10,000 employees, rising turnover prompted this end-to-end analysis using **SQL** and **Power BI**. The analysis identifies key attrition drivers: overtime, business travel, commute distance, low satisfaction scores, and specific departments and job roles with disproportionately high turnover.

Project Objectives



→ Calculate Attrition Rate

Compute total, active, and exited employee counts.

→ Identify High-Risk Segments

Pinpoint job roles, departments, and education levels with highest attrition.

→ Analyze Work Factors

Examine overtime, travel, salary hikes, distance, and tenure.

→ Build Interactive Dashboard

Filterable by Job Role, Job Level, and Gender for HR teams.

Key Metrics Overview

10,000

Total Employees

2,247

Employees Left

7,753

Active Employees

22.47%

Attrition Rate

40

Average Age (yrs)

❏ **1 in 4 employees** has left the organization — a significant retention challenge affecting project delivery and team stability.

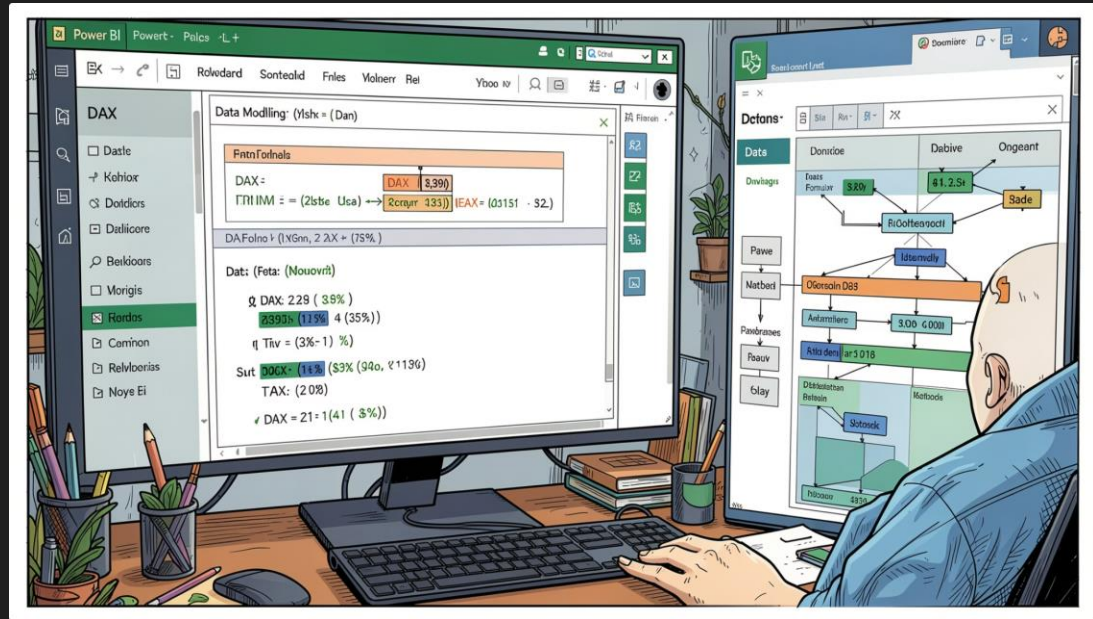
Methodology & Tools

3.1 Data Source – MySQL

The HR dataset was hosted in a MySQL database. Power BI Desktop connected directly via MySQL connector for live employee record queries.

3.2 Dashboard Development – Power BI

- DAX measures for KPIs: Total Employees, Attrition, Attrition Rate
- Bar, donut, line charts, and treemaps for attrition breakdowns
- Cross-filtering slicers: Job Role, Job Level, Gender
- Global Reset button via Power BI bookmarks



DASHBOARD

Top-Level KPI Cards



Five KPI cards provide an at-a-glance workforce summary:

Total Employees

10,000

Total Attrition

2,247

Active Employees

7,753

Attrition Rate

22.47%

Average Age

40 years

Experience Score Distributions

Score distributions (1–5 scale, Job Level 1–8) reveal a substantial proportion of employees rate **Job Involvement**, **Work-Life Balance**, **Job Satisfaction**, and **Job Environment** at the lower end (ratings 1–2), correlating with elevated attrition.

- ① Low engagement scores are a clear signal of culture and satisfaction issues driving turnover.



ATTRITION DRIVERS

Attrition by Travel, Overtime & Commute

Business Travel

Travel Rarely: **1,182 cases (52%)**; No Travel: 375 (16.69%).
Travel frequency alone doesn't fully explain turnover.

Overtime

Overtime employees: **798 cases**; non-overtime: 1,449 (64%). Overtime is a strong individual risk factor despite smaller absolute numbers.

Commute Distance

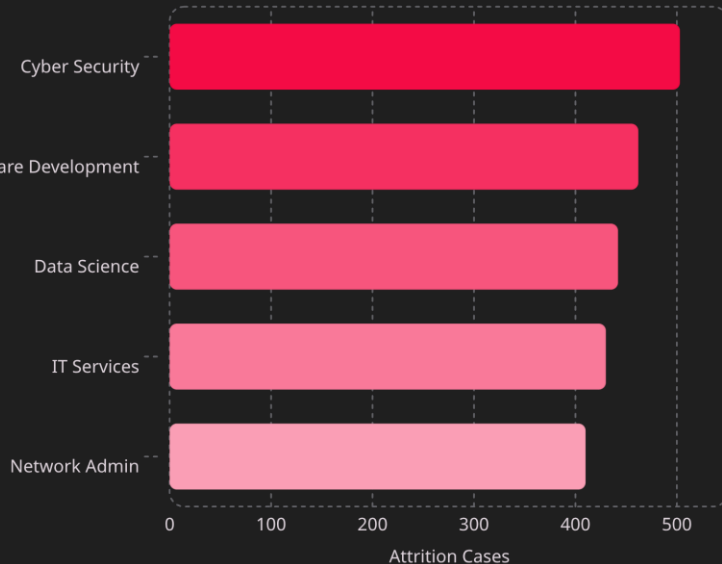
Attrition rises sharply beyond **8 units**, peaking at **61 cases** around 17 units. Longer commutes strongly predict departure.



DEPARTMENTAL BREAKDOWN

Attrition by Department & Education

Department



Department Insights

Cyber Security leads with **503 cases (22.4%)**. The even spread (18–22% each) signals a **systemic, organization-wide issue**.

Education Level

Graduation (702) + Degree (686) = **over 60%** of total attrition, reflecting larger populations and greater external job mobility at these levels.

Key Findings & Insights

→ **22.47% attrition rate**

Nearly 1 in 4 employees has left — a critical retention challenge.

→ **Overtime is a strong driver**

Disproportionately high attrition rate among overtime workers.

→ **4-5 year tenure is a risk window**

Attrition peaks at 227–235 cases, suggesting career stagnation.

→ **Longer commutes increase departure risk**

Attrition rises beyond 8 units, peaking at 61 cases (~17 units).

→ **Low engagement scores correlate with attrition**

Ratings 1–2 in satisfaction, involvement, and work-life balance are overrepresented.

RECOMMENDATIONS

Recommendations & Conclusion

1 Rebalance Overtime

Target Cyber Security & Software Development to reduce burnout-driven turnover.

2 Career Checkpoints

Introduce promotion pathways and lateral movement for employees at the 4–5 year mark.

3 Remote/Hybrid Options

Mitigate commute-related attrition for employees living beyond 8 units.

4 Engagement Surveys

Target employees rating satisfaction and work-life balance at 1–2.

5 Role-Specific Retention

Develop strategies for Consultant and IT roles with elevated attrition rates.

6 Non-Monetary Levers

Combine compensation reviews with recognition, flexibility, and career growth.

7 Recurring Monitoring

Use the Power BI dashboard to track intervention impact over time.



This end-to-end Power BI workflow — from MySQL connection through DAX-driven KPIs to an interactive dashboard — equips HR stakeholders to segment findings by Job Role, Job Level, and Gender and prioritize targeted retention initiatives.